

Introduction to Optimization

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Why should I bother to learn this stuff?

- Many useful engineering models are convex: least squares, energy minimization, resource allocation, economic dispatch, ...
- Recognizing convexity tells you that a numerical solution can often be trusted as a global optimum, not merely a locally good point.
- Convex models can often be solved reliably even when they have many variables and constraints.
- Dual variables can quantify the value of additional resources or relaxed constraints.

Contents

1 Convexity

- Convex sets
- Convex functions
- Properties of convex functions

2 Convex optimization problems

- Convex optimization problems
- First-order optimality conditions
- Duality
- Marginal interpretation of the dual variables

3 Wrap-up



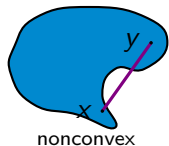
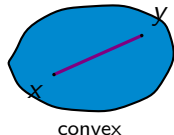
- C is **convex** if for any two points x and y in C the segment $[x, y] \subset C$, i.e.

$$\forall x, y \in C, \forall \theta \in [0, 1], \theta x + (1 - \theta)y \in C.$$

- The **convex hull** of C is the set of **convex combinations** of elements of C , i.e.

$$\text{conv}(C) := \left\{ \sum_{i=1}^K \theta_i x_i \mid K \in \mathbb{N}^*, x_i \in C, \right. \\ \left. \theta_i \in [0, 1] \forall i \in [K], \sum_{i=1}^K \theta_i = 1 \right\}$$

- $\text{conv}(C)$ is the smallest convex set containing C .



Examples of convex sets

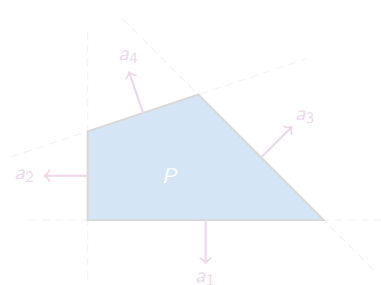
- Linear or affine (sub)-space, half-space, hyperplane.
- Ball (for any norm) and ellipsoid.
- Polyhedron (finite intersection of half-spaces).
- The set of symmetric positive (semi-)definite matrices.

Exercise. Prove these results.

Some sets are not convex:

- The set of integers $\mathbb{Z}$.
- $\{x \in \mathbb{R}^n \mid \|x\|_2 \geq 1\}$.
- The set of orthogonal matrices.

Exercise. Prove these results.



$$P = \{x \in \mathbb{R}^2 \mid a_i^\top x \leq b_i, i \in [4]\}$$

Examples of convex sets

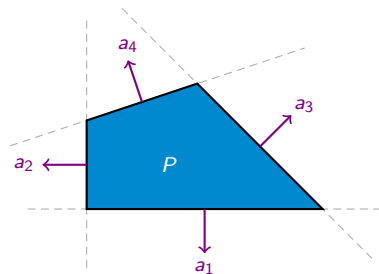
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Operations preserving convexity

Assume that all sets denoted by C (indexed or not) are convex.

- $C_1 + C_2$ and $C_1 \times C_2$ are convex sets.
- For any arbitrary index set $\mathcal{I}$ the intersection $\bigcap_{i \in \mathcal{I}} C_i$ is convex.
- Let f be an affine function. Then $f(C)$ and $f^{-1}(C)$ are convex.
- In particular, $C + x_0$, and tC are convex. The projection of C on any affine space is convex.

Exercise. Prove these results.

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- A function $f : X \rightarrow \bar{\mathbb{R}}$ is **convex** if its epigraph is convex.
- $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex iff

$$\forall t \in [0, 1], \forall x, y \in X, \\ f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

- f is **concave** if $-f$ is convex.

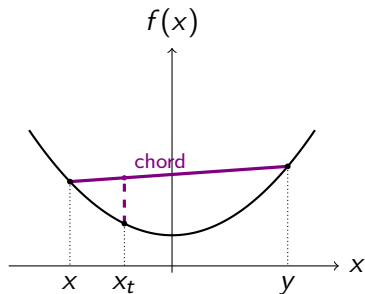


Figure: The graph of a convex function lies below its chords.

Convex function and tangent

Theorem

A differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if it is always above its tangents:

$$f(x) \geq f(y) + \nabla f(y)^\top (x - y) \quad \forall x, \forall y$$

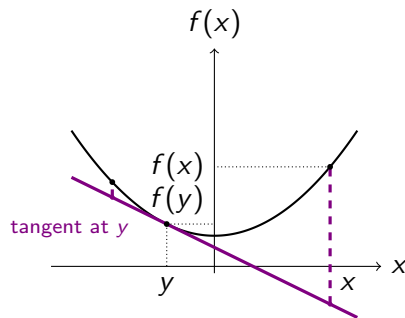


Figure: A differentiable convex function lies above every tangent.

Hessian matrix and semidefinite matrices

- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 function. The **Hessian matrix** of f at x is the matrix of second derivatives

$$\nabla^2 f(x) := \begin{pmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{pmatrix}.$$

- $\nabla^2 f(x)$ is symmetric.
- A symmetric matrix A is
 - ▶ positive semidefinite ($A \succcurlyeq 0$) if $\forall z \in \mathbb{R}^n, z^\top A z \geq 0$
 - ▶ positive definite ($A \succ 0$) if $\forall z \in \mathbb{R}^n \setminus \{0\}, z^\top A z > 0$
- If A is symmetric, then:
 - ▶ A is **positive semidefinite** iff all its eigenvalues are nonnegative (≥ 0),
 - ▶ A is **positive definite** iff all its eigenvalues are positive (> 0).
 - ▶ $A - \alpha I$ is positive semidefinite (also written $A \succcurlyeq \alpha I$) iff all its eigenvalues are $\geq \alpha$.

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Convex functions: strict and strong convexity

- $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is strictly convex iff

$$\forall t \in]0, 1[, \quad \forall x, y \in X, \quad f(tx + (1-t)y) < tf(x) + (1-t)f(y)$$

- $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is α -convex iff $\forall t \in]0, 1[, \quad \forall x, y \in X,$

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - \frac{1}{2}\alpha t(1-t)\|x - y\|^2$$

- If $f \in C^1(\mathbb{R}^n)$

- ▶ $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$ iff f convex
- ▶ if strict inequality holds, then f strictly convex
- ▶ f is α -convex iff $\forall x, y \in \mathbb{R}^n$

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- If $f \in C^2(\mathbb{R}^n),$

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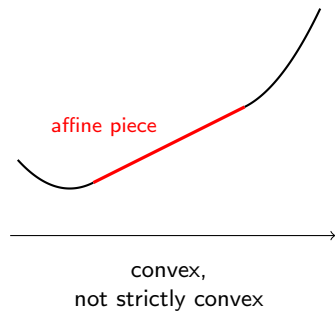
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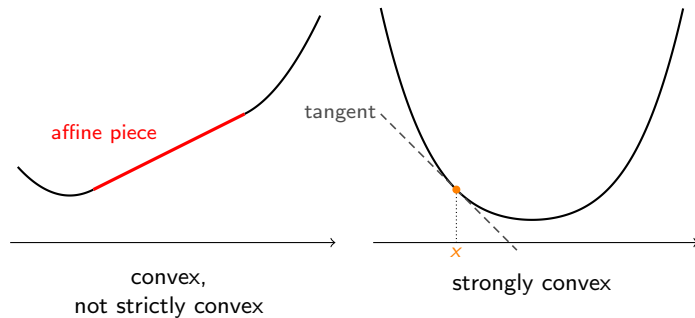
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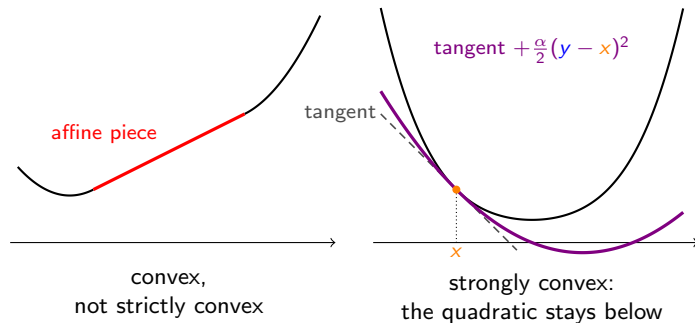
Strict and strong convexity: pictures



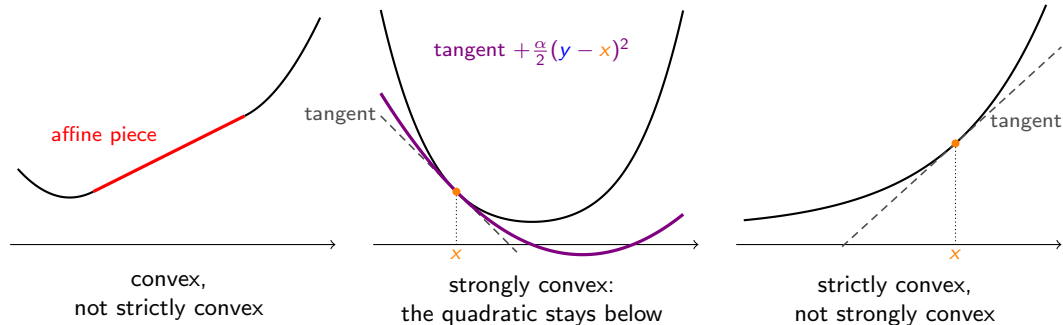
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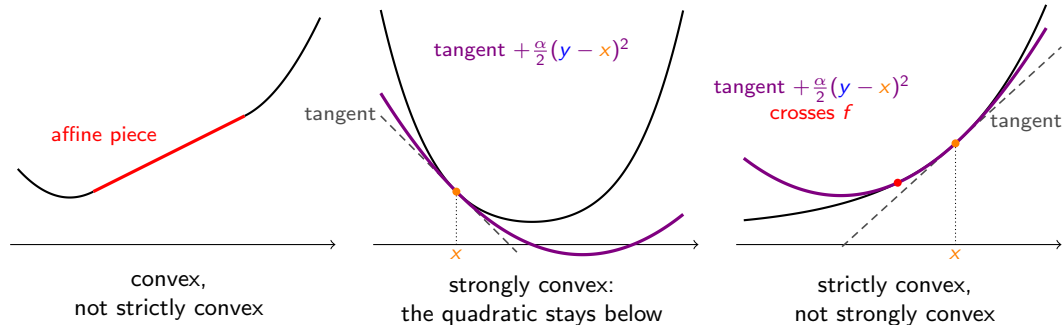
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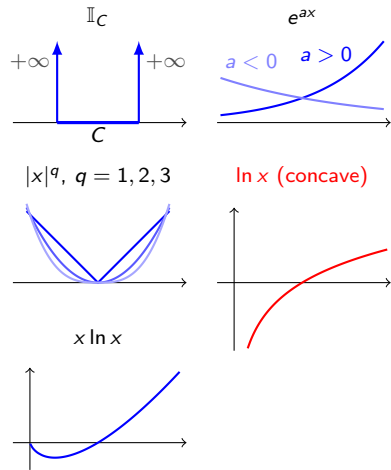
Important examples

- The **indicator function** of a set $C \subset X$,

$$\mathbb{I}_C(x) := \begin{cases} 0 & \text{if } x \in C \\ +\infty & \text{otherwise} \end{cases}$$

is convex iff C is convex.

- $x \mapsto e^{ax}$ is convex for any $a \in \mathbb{R}$
- $x \mapsto \|x\|^q$ is convex for $q \geq 1$ and any norm
- $x \mapsto \ln(x)$ is concave on $]0, +\infty[$
- $x \mapsto x \ln(x)$ is convex on $]0, +\infty[$



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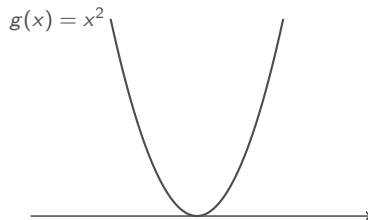
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3 Wrap-up

Basic properties

- If f, g convex, $t > 0$, then $tf + g$ is convex.
- **⚠** If f, g convex, then $f \circ g$ not necessarily convex, but:
 - ▶ If f convex non-decreasing, g convex, then $f \circ g$ convex.
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- If $(f_i)_{i \in I}$ is a family of convex functions, then $\sup_{i \in I} f_i$ is convex.
- The domain and the sublevel sets of a convex function are convex.

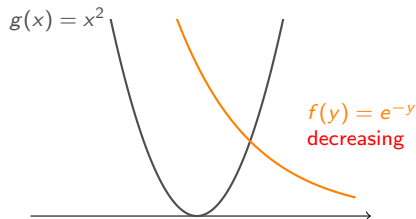


Exercise. Prove these results

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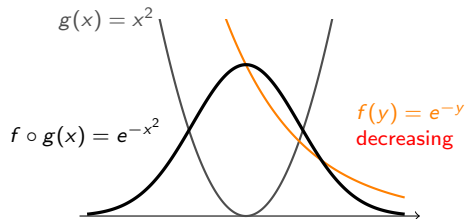
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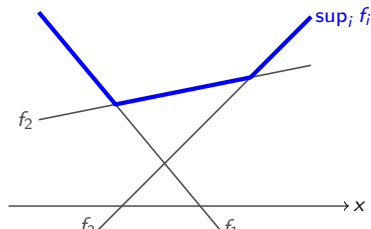
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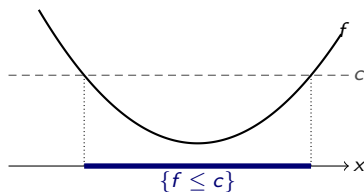
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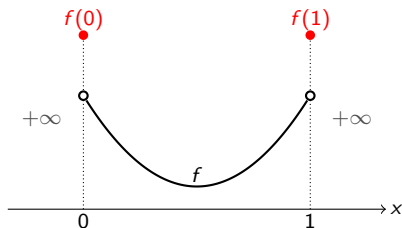
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Convex function: regularity



Consider a convex function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$.

- f is continuous (on $\mathbb{R}^n$) if and only if $\text{dom}(f) = \mathbb{R}^n$ (i.e., if it is finite everywhere)
- f is continuous on the interior of its domain
- f is Lipschitz continuous on any compact subset of the interior of its domain



Convex, with $\text{dom}(f) = [0, 1]$: continuous on $]0, 1[$, but not at 0 and 1 .

Convex functions: existence, uniqueness and characterization of minima ♡

- If f is convex, then any local minimum is a global minimum. Further, the set of global minimizers is convex.
- If f is strictly convex, then there exists at most one global minimizer.
- If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is α -convex, then it admits a unique global minimizer.

Exercise. Prove these results.

First-order optimality conditions are necessary and sufficient in the convex case:

Theorem

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and differentiable. Then $x^\#$ is a global minimizer of f iff $\nabla f(x^\#) = 0$.

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First-order optimality on a convex set

Let $X \subseteq \mathbb{R}^n$ be convex and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and differentiable.

Theorem

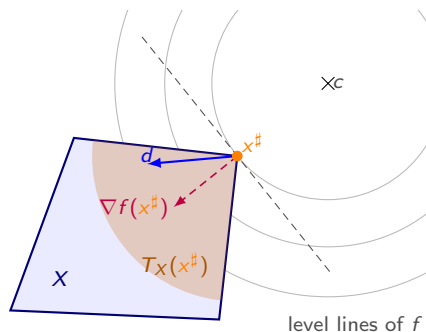
A point $x^\# \in X$ is a global minimizer of

$$\min_{x \in X} f(x)$$

if and only if

$$\nabla f(x^\#)^\top d \geq 0 \quad \forall d \in T_X(x^\#).$$

Thus, for convex problems, the constrained first-order necessary condition from Lecture 1 is also sufficient.



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Abstract convex optimization problems

An abstract optimization problem of the form

$$\underset{x \in X}{\text{Min}} \quad f(x)$$

is **convex** if $f : X \rightarrow \overline{\mathbb{R}}$ is convex and X is a convex set.

A convex optimization problem has nice properties:

- any local minimum is a global minimum,
- if f is strictly convex, there exists at most one global minimizer,
- if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is α -convex and X is nonempty, closed and convex, there exists a unique global minimizer,
- if f is differentiable, first-order optimality conditions are necessary and sufficient.

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Differentiable convex optimization problem

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{Min}} & f(x) \\ \text{s.t.} & g_i(x) = 0 \quad \forall i \in [n_E] \\ & h_j(x) \leq 0 \quad \forall j \in [n_I] \end{array} \quad (P)$$

This is called a **convex differentiable** optimization problem if f is convex differentiable, each g_i is affine, and each h_j is convex differentiable.

Exercise. Show that a convex differentiable optimization problem is a convex optimization problem.

Why bother with convex optimization?

- Many applications:
 - ▶ Portfolio optimization
 - ▶ Statistical regression
 - ▶ Model fitting
 - ▶ Optimal / Robust control
 - ▶ Signal processing
 - ▶ ...
- Strong theoretical results:
 - ▶ KKT are necessary and sufficient (no local optima or saddle point)
 - ▶ Strong duality theory
 - ▶ Complexity theory
- Efficient algorithms in theory and practice
 - ➡ Algorithms can be applied to other settings in a more or less heuristic way

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Theorem (KKT - convex case)

Consider a **convex** differentiable optimization problem

$$\text{Min}_{\tilde{\mathbf{x}} \in \mathbb{R}^n} \left\{ f(\tilde{\mathbf{x}}) \mid g_i(\tilde{\mathbf{x}}) = 0, \forall i \in [n_E], \quad h_j(\tilde{\mathbf{x}}) \leq 0, \forall j \in [n_I] \right\}$$

If the constraints are qualified^a at $\mathbf{x}$, then $\mathbf{x}$ is optimal **if and only if** there exists dual variables λ, μ such that

$$\left\{ \begin{array}{ll} \nabla f(\mathbf{x}) + \sum_{i=1}^{n_E} \lambda_i \nabla g_i(\mathbf{x}) + \sum_{j=1}^{n_I} \mu_j \nabla h_j(\mathbf{x}) = 0 & \nabla_x \mathcal{L} = 0 \\ g(\mathbf{x}) = 0, \quad h(\mathbf{x}) \leq 0 & \text{Primal feasibility} \\ \lambda \in \mathbb{R}^{n_E}, \quad \mu \in \mathbb{R}_+^{n_I} & \text{dual feasibility} \\ \mu_j h_j(\mathbf{x}) = 0 \quad \forall j \in [n_I] & \text{complementary slackness} \end{array} \right.$$

^aEven without qualification, in the convex case, any primal-dual pair satisfying KKT is optimal.

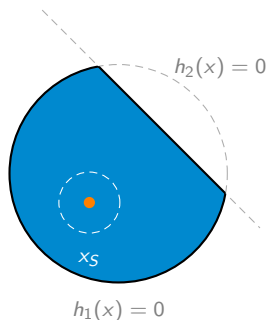
Slater qualification condition

$$\begin{array}{ll} \text{Min}_{x \in \mathbb{R}^n} & f(x) \\ \text{s.t.} & g_i(x) = 0 \quad \forall i \in [n_E] \\ & h_j(x) \leq 0 \quad \forall j \in [n_I] \end{array} \quad (P)$$

where f is convex differentiable, each g_i is affine, and each h_j is convex differentiable.

We say that $x_S \in \mathbb{R}^n$ is a **Slater point** if it strictly satisfies the inequality constraints, i.e.

- $g_i(x_S) = 0$ for all $i \in [n_E]$,
- $h_j(x_S) < 0$ for all $j \in [n_I]$.



strictly feasible: $h_j(x_S) < 0$ for all j .

Theorem

If there exists a **Slater point** x_S , then the constraints are qualified at every feasible point.

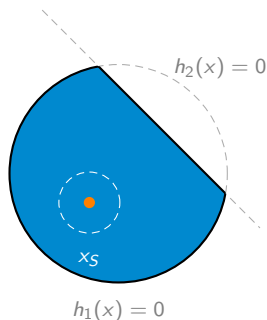
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Contents

1 Convexity

- Convex sets
- Convex functions
- Properties of convex functions

2 Convex optimization problems

- Convex optimization problems
- First-order optimality conditions
- **Duality**
- Marginal interpretation of the dual variables

3 Wrap-up

Lagrangian duality — recall

For the primal problem

$$\begin{aligned} (P) \quad & \text{Min}_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \\ & \text{s.t.} \quad g_i(\mathbf{x}) = 0 \quad \forall i \in [n_E], \\ & \quad \quad h_j(\mathbf{x}) \leq 0 \quad \forall j \in [n_I], \end{aligned}$$

recall the Lagrangian and the dual function

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\mathbf{x}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x}) + \boldsymbol{\mu}^\top \mathbf{h}(\mathbf{x}),$$

$$d(\boldsymbol{\lambda}, \boldsymbol{\mu}) := \inf_{\mathbf{x} \in \mathbb{R}^n} \mathcal{L}(\mathbf{x}; \boldsymbol{\lambda}, \boldsymbol{\mu}),$$

for $(\boldsymbol{\lambda}, \boldsymbol{\mu}) \in \mathbb{R}^{n_E} \times \mathbb{R}_+^{n_I}$.

The dual problem is

$$(D) \quad \text{Max}_{\boldsymbol{\lambda} \in \mathbb{R}^{n_E}, \boldsymbol{\mu} \in \mathbb{R}_+^{n_I}} d(\boldsymbol{\lambda}, \boldsymbol{\mu}),$$

and weak duality gives

$$d(\boldsymbol{\lambda}, \boldsymbol{\mu}) \leq \text{val}(D) \leq \text{val}(P) \leq f(\mathbf{x})$$

for every primal-feasible $\mathbf{x}$ and dual-feasible $(\boldsymbol{\lambda}, \boldsymbol{\mu})$.

Question: when does weak duality become equality?

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Strong duality and KKT conditions



If (P) is **convex** and there exists a Slater point, then

$$\text{val}(P) = \text{val}(D),$$

i.e.

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Theorem

If (P) is convex and there exists a Slater point, then for $x^\# \in \mathbb{R}^n$ and $(\lambda^\#, \mu^\#) \in \mathbb{R}^{n_E} \times \mathbb{R}_+^{n_I}$ the following assertions are equivalent:

- 1 $x^\#$ is an optimal solution of (P) and $(\lambda^\#, \mu^\#)$ is an optimal solution of (D) ,
- 2 $(x^\#; \lambda^\#, \mu^\#)$ is a saddle point of $\mathcal{L}$,
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Marginal interpretation of the dual variables



Consider the perturbed value function

$$\begin{aligned} v(q) = \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & g(x) = 0, \\ & h(x) \leq q. \end{aligned}$$

In particular, $v(0) = \text{val}(P)$.

Assume that (P) is convex and satisfies Slater's condition, and let $(\lambda^\#, \mu^\#)$ be optimal dual variables at $q = 0$. Then

$$v(q) \geq v(0) - (\mu^\#)^\top q.$$

Thus $-\mu^\#$ gives the marginal sensitivity of the optimal value to the right-hand side. In particular, if v is differentiable at 0,

$$\nabla_q v(0) = -\mu^\#.$$

Exercise: Prove this result.

Marginal interpretation of the dual variables



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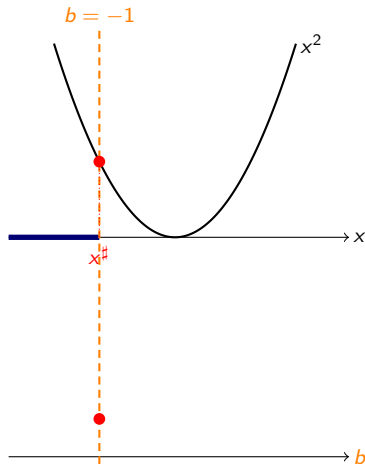
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$$\begin{array}{ll} \text{Min}_{x \in \mathbb{R}} & x^2 \\ \text{s.t.} & x \leq b \end{array}$$

- 1 Do optimal dual variables exist?
- 2 Without solving the dual, give the optimal dual variable μ_b .

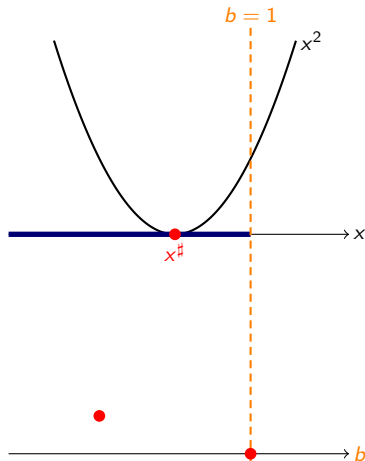


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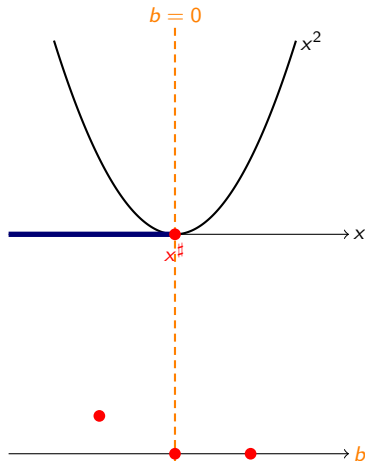


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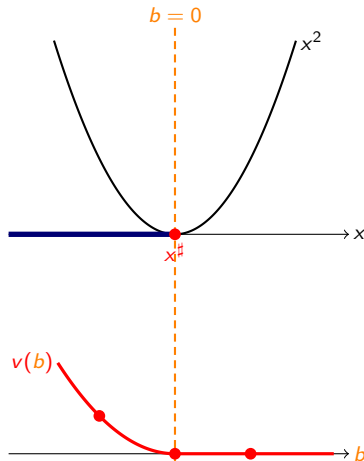


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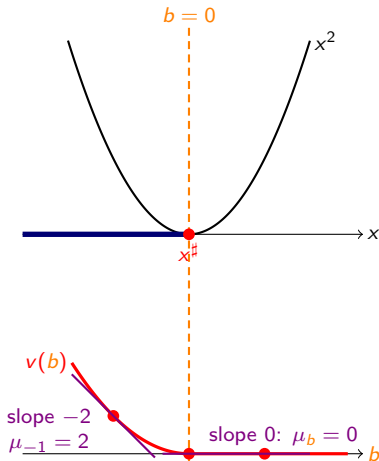


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 (P) \quad & \underset{x \in \mathbb{R}^n}{\text{Min}} \quad f(x) \\
 & \text{s.t.} \quad Ax = b \\
 & \quad \quad h_j(x) \leq 0 \quad \forall j \in [n_I]
 \end{aligned}$$

with associated Lagrangian

$$\mathcal{L}(x; \lambda, \mu) := f(x) + \lambda^\top (Ax - b) + \mu^\top h(x)$$

In the convex case, for $\mu \geq 0$, the map $x \mapsto \mathcal{L}(x; \lambda, \mu)$ is convex. The KKT conditions can therefore be seen as:

- ① $\nabla_x \mathcal{L}(x; \lambda, \mu) = 0$ (Lagrangian minimized in x)
- ② $Ax = b, h(x) \leq 0$ (x primal feasible; $Ax = b$ also follows from $\nabla_\lambda \mathcal{L} = 0$)
- ③ $\mu \geq 0$ ((λ, μ) dual feasible)
- ④ $\mu_j = 0$ or $h_j(x) = 0$, for all $j \in [n_I]$

(complementary slackness $\leadsto 2^{n_I}$ possibilities).



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What you should know

- What a convex set and a convex function are, and the main operations preserving convexity.
- The distinction between convex, strictly convex and strongly convex functions.
- The consequences of convexity for global optimality, existence and uniqueness.
- For convex problems, KKT conditions become sufficient; under Slater's condition they characterize optimality and strong duality holds.
- The marginal interpretation of optimal dual variables.

What you should be able to do

- Show that a simple set or function is convex.
- Recognize whether an optimization problem is convex.
- Use strict or strong convexity to establish uniqueness or existence of a solution.
- Interpret a dual multiplier as a marginal value.